

On the Faithfulness of Post-Hoc Concept Bottleneck Models

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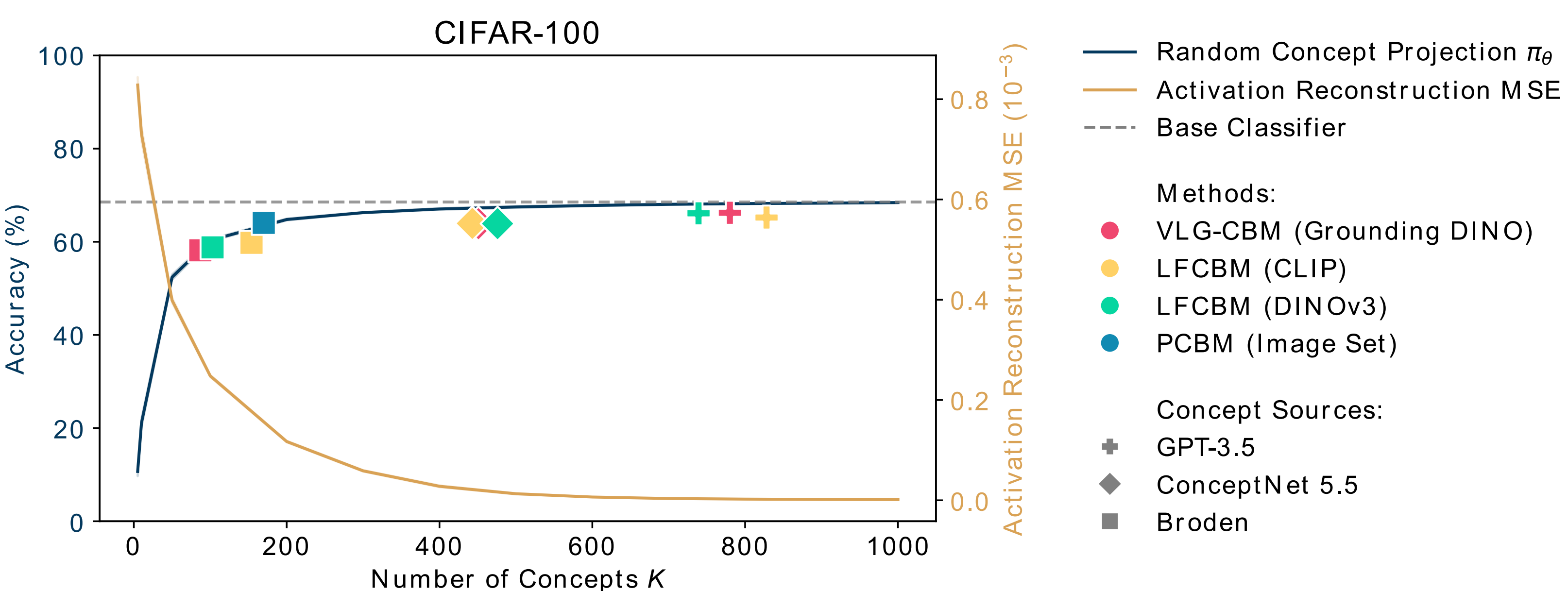
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The Problem

Post-Hoc Concept Bottleneck Models (Post-Hoc CBMs) map frozen backbone features to human concepts before classification. They often achieve strong downstream accuracy. However, high performance is possible with meaningless concepts extracted by random projections. Thus, task accuracy alone is not sufficient evidence for faithful concepts.



Faithfulness

Definition 1. We define a concept projection π_θ as faithful if it minimizes the expected risk over the true task distribution $P_{\text{task}}(x, c)$:

$$\arg \min_{\theta} \mathbb{E}_{(x,c) \sim P_{\text{task}}} [\mathcal{L}(\pi_\theta(f(x)), c)], \quad (1)$$

for some loss function $\mathcal{L}$ measuring the alignment of the predicted concepts $\pi_\theta(f(x))$ and the ground truth concepts c for an input x .

Accuracy Does Not Imply Faithfulness

Distribution shifts degrade the faithfulness of concept projection more than downstream accuracy indicates (results for the Elements dataset [1]).

Unfaithfulness
Projections trained on data with a distribution shift fail to detect the task concepts.

Misleading Accuracy
The classifier trained on the learned concept projection always achieves high accuracy.

Method	Dataset/ Label	Concept Projection π_θ			Downstream Classifier	
		$\text{sim}_{\triangleleft}$ $\uparrow$	Acc. P_{aux} $\uparrow$	Acc. P_{task} $\uparrow$	Acc. h $\uparrow$	Acc. h^* $\uparrow$
<i>(1) Training on Auxiliary Distributions (Oracle Labels)</i>						
PCBM [2]	P_{task}	0.96 $\pm$ 0.01	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00	1.00 $\pm$ 0.00
	$P_{\text{near}}^{\forall}$	0.97 $\pm$ 0.01	1.00 $\pm$ 0.00	0.70 $\pm$ 0.02	1.00 $\pm$ 0.00	0.99 $\pm$ 0.01
	$P_{\text{near}}^{\exists}$	0.57 $\pm$ 0.02	0.82 $\pm$ 0.02	0.89 $\pm$ 0.01	1.00 $\pm$ 0.00	0.85 $\pm$ 0.04
	P_{OOD}	0.47 $\pm$ 0.01	0.98 $\pm$ 0.01	0.77 $\pm$ 0.01	1.00 $\pm$ 0.00	0.21 $\pm$ 0.05
<i>(2) Surrogate Concept Labels (Training Dist.: P_{task})</i>						
LFCBM [3]	25% Label Noise	0.02 $\pm$ 0.03	0.75 $\pm$ 0.01	0.98 $\pm$ 0.02	1.00 $\pm$ 0.00	0.98 $\pm$ 0.03
	CLIP [4]	0.05 $\pm$ 0.02	0.79 $\pm$ 0.01	0.56 $\pm$ 0.02	0.99 $\pm$ 0.00	0.54 $\pm$ 0.04
	DINOv3 [5]	0.05 $\pm$ 0.01	0.85 $\pm$ 0.01	0.58 $\pm$ 0.02	1.00 $\pm$ 0.00	0.78 $\pm$ 0.05
VLG-CBM [6]	G-DINO [7]	0.02 $\pm$ 0.02	0.74 $\pm$ 0.01	0.85 $\pm$ 0.01	1.00 $\pm$ 0.01	0.04 $\pm$ 0.01

If we want to evaluate whether a concept projection is faithful, we have to consider the concept projection in the downstream domain. If we do not do this, we may violate Definition 1.

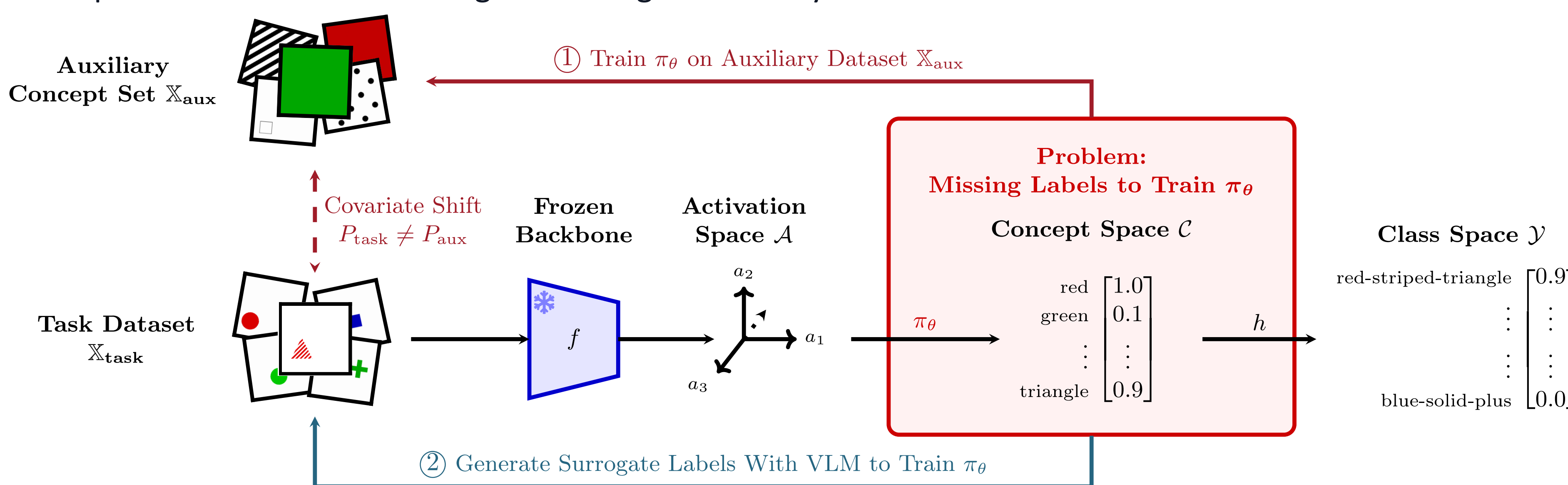
Ground Truth
The classifier trained on the ground truth concept projection does not.

Can You Trust Post-Hoc Concept Bottleneck Models?

Distribution shifts in the concept training data can lead to unfaithful bottlenecks.

Potential Distribution Shifts

We want to learn the distribution $P_{\text{task}}(x, c) = P_{\text{task}}(x)P_{\text{task}}(c|x)$. But in practice, concept labels are often unavailable in the task domain. Methods therefore learn the concept projection either ① from an auxiliary concept dataset or ② from surrogate labels generated by a VLM in the downstream domain.



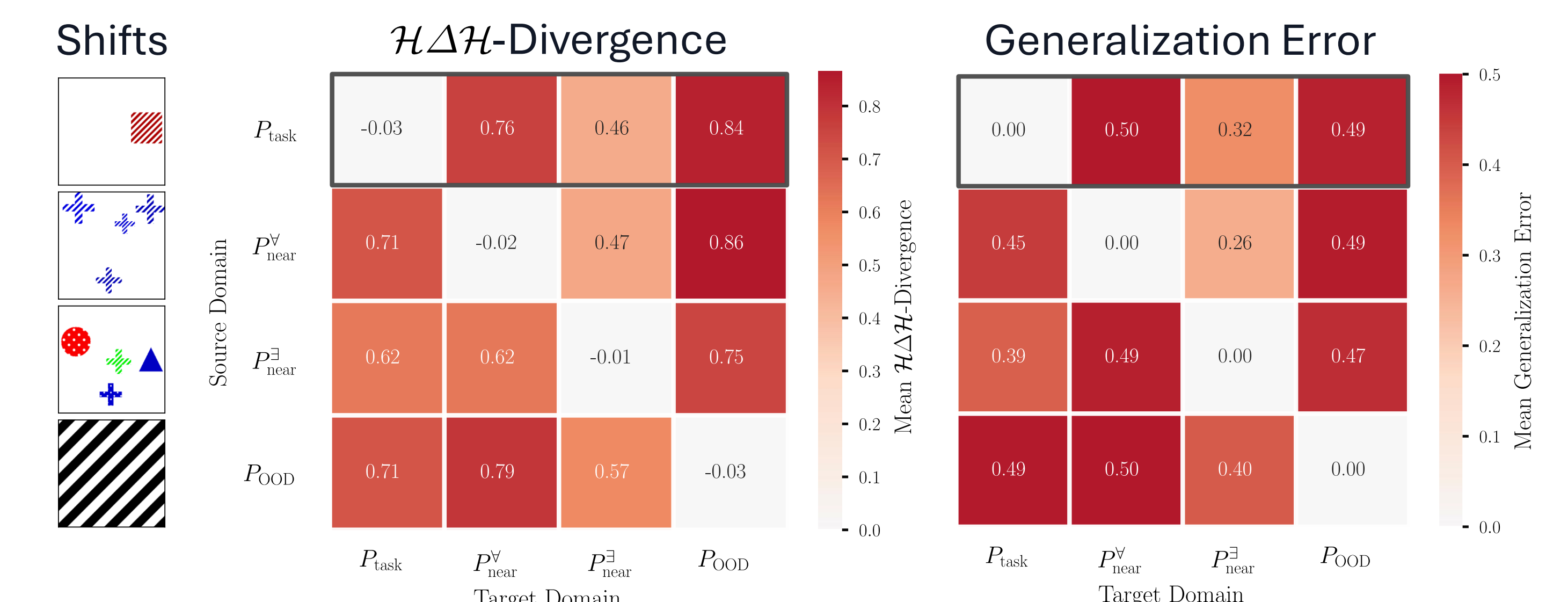
① **Auxiliary concept data shift:** The concept projection is learned on an auxiliary concept dataset whose input distribution differs from the task distribution, leading to a covariate shift.

② **Surrogate label shift:** The concept projection is learned from VLM-generated labels on the task data, where systematic labeling errors can bias the learned concepts.

How Can We Measure Distribution Shifts?

① **Auxiliary concept data shift ($P_{\text{aux}}(x) \neq P_{\text{task}}(x)$):**

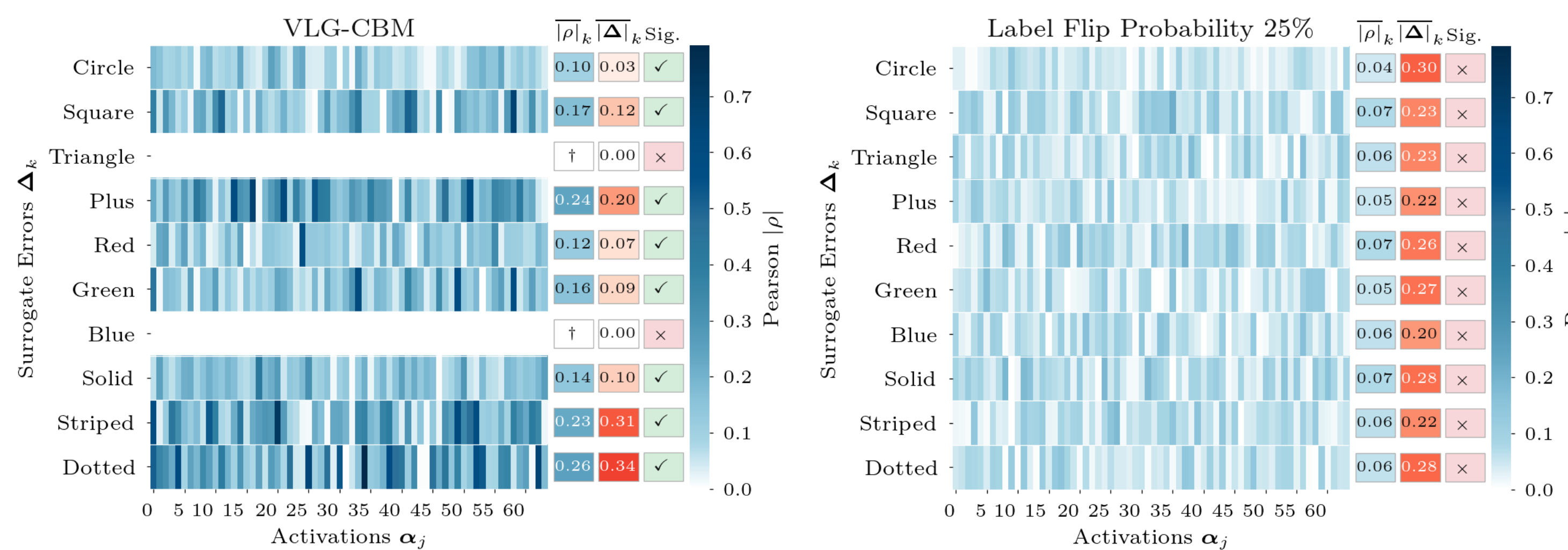
Even with the same semantic concept definitions, a covariate shift between an auxiliary and the task distribution can make the learned projection unfaithful.



We show based on [8] that $\mathcal{H}\Delta\mathcal{H}$ -divergence can quantify this shift by training a domain classifier to distinguish both domains. If the two domains are easy to separate, the divergence is large, indicating that trained concept projections will not be faithful.

② **Surrogate-label shifts ($P_{\text{task}}(\tilde{c}|x) \neq P_{\text{task}}(c|x)$):**

Even with the same task data distribution, a shift in the concept labels can make the projection unfaithful. Random noise may cancel out, but not systematic errors.



We show that strong correlations ($|\rho|_{kj}$) between surrogate label errors (Δ_k) and backbone activations (α_j) lead to unfaithfulness. In this case, the concept projection may learn the VLM's bias instead of the intended concepts.

Takeaways

© High accuracy does not imply faithful concepts projections.

⚠ Distribution shifts lead to unfaithful concepts projections.

☑ Concept projections should be evaluated directly.



Project Page
posthoc-cbm-faithfulness.github.io

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